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A Variable Switching Frequency Control Technique for DC-AC Flying Capacitor Multilevel Converters to Improve Efficiency and Inductor Utilization

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***Abstract*—This work proposes a variable switching frequency scheme for a flying capacitor multilevel (FCML) converter in dc-ac operation to better utilize the output inductor for its rated peak-to-peak current ripple and substantially reduce converter losses over the ac line cycle. The proposed technique holds the inductor current ripple constant by leveraging the duty-cycle dependent inductor current ripple, and variably changes the switching frequency to hold this constant at its rated design value. Practical lower switching limits are taken into consideration in developing this approach, and experimental results validate the feasibility in a common micro-controller framework. Hardware results are provided for a 400 V_{dc} to 240 V_{rms} 6-level FCML inverter and showcase loss reductions on the order of $10-35\%$ across the measured power range, compared to conventional control with fixed frequency operation.**

***Index Terms*—dc-ac converters, multilevel converters, modulation, hybrid switched-capacitor converters**

I. Introduction

The increased adoption of hybrid switched-capacitor and multilevel converter topologies such as the FCML converter is a direct result of the comparative passive component volume reduction and increased efficiency they boast. This is accomplished via the use of high figure-of-merit low-voltage switches [1]–[4], and use of energy dense capacitors [5]. For dc-ac applications such as renewable energy conversion or electric vehicle charging, where efficiency and power-density are critical design objectives, the FCML converter is an attractive solution [6].

While the FCML converter leverages the power density of capacitors to reduce overall converter volume, the inductor remains a large and lossy component in the circuit, owing to its comparatively poor energy density. In dc-ac operation, the inductor is typically sized to limit the current ripple at a set switching frequency f_{sw}. However, as the peak-to-peak inductor current ripple $\Delta i_{L,\text{pp}}$ is a strong function of the duty cycle in the dc-ac FCML converter, the resulting converter has over-rated passive components at many operating conditions, and increased losses at a higher than needed switching frequency. Under constant f_{sw}, the inductor volt-seconds vary over the line cycle. Thus to meet a maximum current ripple specification, the inductance and constant f_{sw} must be chosen for the worst point in the line cycle.

Previous work in the dc-dc space has investigated modulation techniques for inductor current ripple control in the multilevel buck-boost transition region as well as modulation schemes to change the effective level count of the FCML [7], [8]. For dc-ac FCML converters, variable frequency control has been specifically used to limit the voltage ripple on flying capacitors [9], [10], and to mitigate capacitor ripple during overload conditions [11].

This work develops a set of guidelines and control for a variable switching frequency scheme designed for a constant inductor current ripple $\Delta i_{L,\text{pp}}$ for an FCML converter operated in dc-ac mode. The required switching frequency is derived based on the dependency of the inductor current ripple on the instantaneous duty cycle and switching frequency. Further constraints are set for the lower switching frequency limits based on the allowable flying capacitor voltage ripple $\Delta v_{C,\text{fly}}$, and the LC filter on the output of the converter. These limits create a set of guidelines that ensure that all passive ripple conditions are achieved, even when the switching frequency is reduced beyond its nominal rating. The overall reduction in switching frequency enables the realization of higher efficiencies by reducing switching-related losses.

This paper continues as follows: Section II explores the general guidelines for developing the variable switching frequency scheme, Section III presents a hardware validation with substantial efficiency improvements compared to conventional techniques, and Section IV provides concluding remarks.

II. Overview of Theory and Control

A. Conventional Operation

An N-level FCML converter is composed of $N-1$ switch pairs and $N-2$ flying capacitors. Each flying capacitor has a voltage with a nominal dc offset of $v_{C,k} = V_{\text{in}} \cdot k/(N-1)$, where $k \in [1, N-2]$. Each switch must be rated to block a nominal dc voltage of $\frac{V_{\text{in}}}{N-1}$. Under conventional operation with phase-shifted pulse-width modulation (PS-PWM) [12],

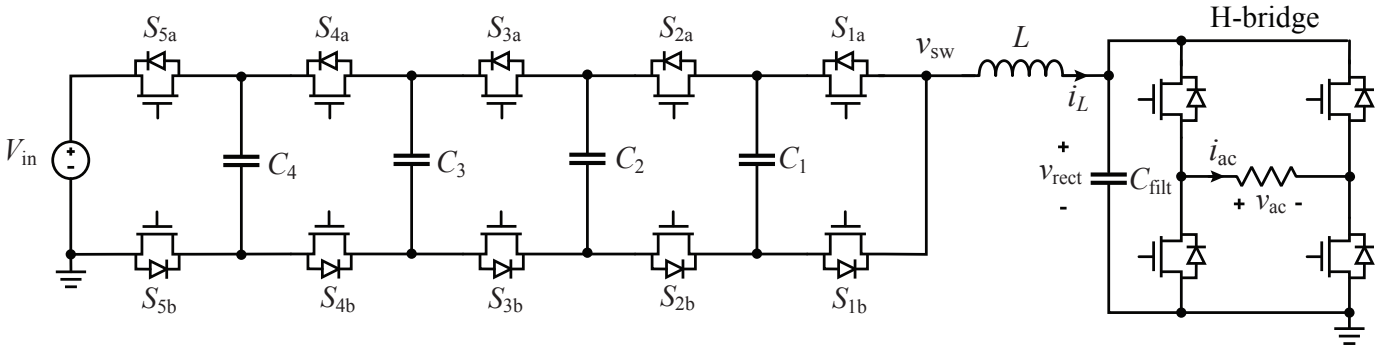


Fig. 1: A 6-level FCML converter with an H-bridge unfolder for dc-ac operation.

the effective frequency observed at the switch node v_{sw} is given by

$$f_{eff} = (N-1) \cdot f_{sw}. \tag{1}$$

For a given switching frequency and inductance value, the resulting inductor current ripple is reduced on the order of $(N-1)^2$ in comparison to a conventional buck converter and is described by

$$\Delta i_{L,pp} = \frac{V_{in} D_{eff}(1-D_{eff})}{L f_{sw}(N-1)^2}, \tag{2}$$

where the effective duty cycle at the switch node is given by

$$D_{eff} = D(N-1) - \lfloor D(N-1) \rfloor. \tag{3}$$

Fig. 1 shows a 6-level FCML converter with an H-bridge unfolder at the output used to convert the rectified sinusoid at the switch node into the ac output waveforms. The duty cycle over a half-line cycle used to generate the ac output is shown in Fig. 2a.

Subsequently, for a dc-ac FCML converter with an unfolder,

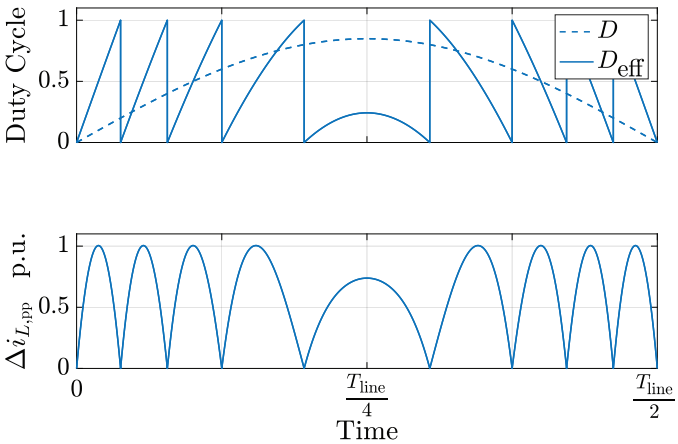


Fig. 2: a) Duty cycle D and corresponding D_{eff} across half of a line cycle. b) Normalized inductor current ripple at each duty cycle for a 6-level FCML inverter during half of a line cycle showing the impact of the duty cycle D on inductor current ripple.

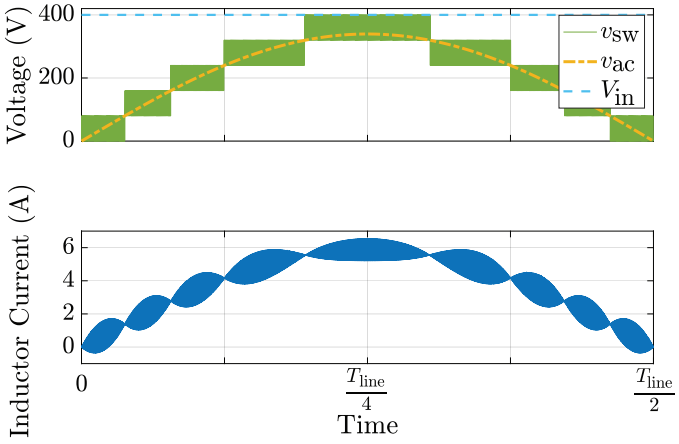


Fig. 3: Calculated waveforms at $V_{in} = 400\,\mathrm{V_{dc}}$, $v_{ac} = 240\,\mathrm{V_{rms}}$, $P_{out} = 1\,\mathrm{kW}$, $L = 22\,\mu\mathrm{H}$, $f_{sw} = 100\,\mathrm{kHz}$.

the inductor current ripple $\Delta i_{L,pp}$ varies since the duty cycle is a rectified sinusoid. At certain duty cycles $D = \frac{k}{N-1}$, $k \in [1, N-2]$, the effective duty cycle D_{eff} is 0, and $\Delta i_{L,pp} = 0$ A as per the expression in (2). Fig. 2b also shows the normalized inductor current ripple across half of a line-cycle [**?**], and how D_{eff} corresponds with $\Delta i_{L,pp} = 0\,\mathrm{A}$ around these nominal duty cycles.

While the inductor in an FCML inverter might be sized for the ripple at the peak of the line cycle, the maximum inductor current ripple occurs when $D_{eff} = \frac{1}{2}$ and is calculated as

$$\Delta i_{L,pp,\,max} = \frac{V_{in}}{4 L f_{sw}(N-1)^2}. \tag{4}$$

Thus, with regards to this rated current ripple, the resulting switching frequency f_{sw} is higher than necessary around the low-ripple duty cycles. This results in unnecessarily high losses, particularly in the switches.

B. Variable Switching Frequency

By rearranging the expression in (2), it is possible to derive an expression to achieve a fixed inductor current ripple. This work proposes varying the switching frequency along the line

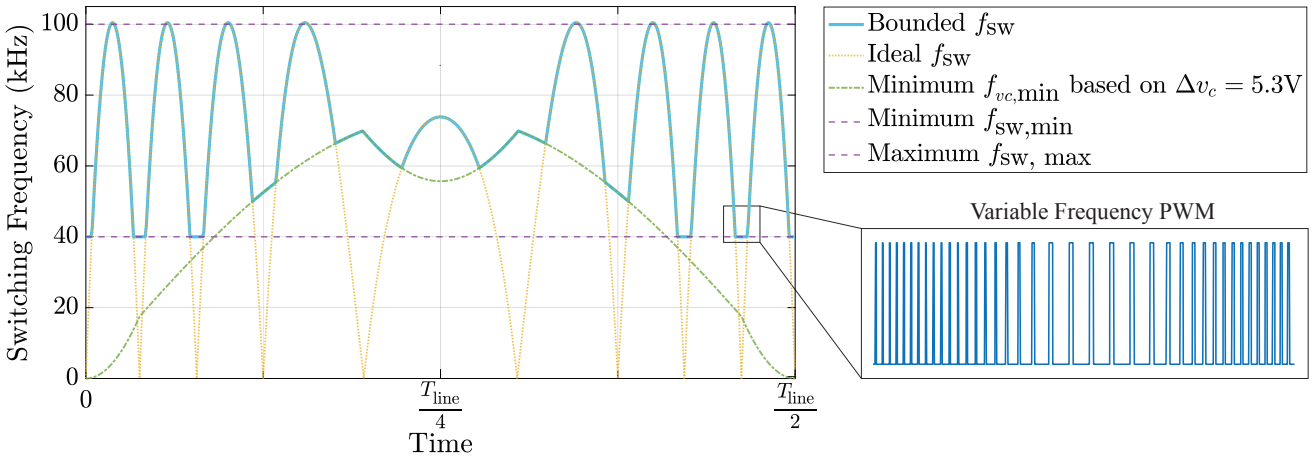


Fig. 4: Variable switching frequency f_{sw} showing the minimum constraints set by the capacitor voltage ripple and the LC filter. Also shown are the PWM signals for the FCML inverter near the minimum switching frequency region.

cycle using (5) to reduce converter losses. Here, the effective duty cycle D_{eff} is a time-varying function of the ac duty cycle.

$$f_{\text{sw}}(t) = \frac{V_{\text{in}} D_{\text{eff}}(1 - D_{\text{eff}})}{\Delta i_{L,\text{pp, max}} L(N-1)^2} \tag{5}$$

There are converter operating constraints that dictate a lower limit of the switching frequency. First, the peak-to-peak flying capacitor voltage ripple $\Delta v_{C,\text{fly}}$ must be limited, as this ripple appears across the adjacent switch pair and must be sufficiently small so as to not cause over-voltage conditions. The ripple $\Delta v_{C,\text{fly}}$ is usually sized as a fraction of the nominal switch blocking voltage $\frac{V_{\text{in}}}{N-1}$ [13].

Assuming a fixed maximum allowable voltage ripple $\Delta v_{C,\text{max}}$, a lower switching frequency limit $f_{v_C,\text{ min}}$ is derived as a function of the flying capacitance C_{fly}, and the instantaneous load current i_{ac}. The function $f_{v_C,\text{ min}}$ in (6) is piece-wise, as the the flying capacitor voltage ripple has different duty-cycle dependent charging times due to PS-PWM operation [13]–[15].

$$f_{v_C,\text{ min}} = \begin{cases} \frac{|i_{\text{ac}}| \cdot D_{\text{eff}}}{(N-1)\cdot\Delta v_{C,\text{max}}\cdot C_{\text{fly}}} & D \le \frac{1}{N-1} \\ \frac{|i_{\text{ac}}|}{(N-1)\cdot\Delta v_{C,\text{max}}\cdot C_{\text{fly}}} & \frac{1}{N-1} < D < \frac{N-2}{N-1} \\ \frac{|i_{\text{ac}}|\cdot(1-D_{\text{eff}})}{(N-1)\cdot\Delta v_{C,\text{max}}\cdot C_{\text{fly}}} & \frac{N-2}{N-1} \le D \end{cases} \tag{6}$$

Second, an absolute lower switching frequency $f_{\text{sw,min}}$ is set based on the LC filter at the output. The effective switching frequency at the switch node is f_{eff}, and must be sufficiently higher than the corner frequency of the LC filter given by $f_{\text{corner}} = \frac{1}{2\pi\sqrt{LC_{\text{filt}}}}$ to avoid resonant behavior. For a given converter, L and C_{filt} are sized to meet a predefined output voltage ripple rating [3], [11], [16], and to achieve sufficient attenuation of f_{eff}. Thus, the switching frequency f_{sw} needs to satisfy the expression described by

$$f_{\text{sw, min}} = \alpha_{LC} \frac{1}{(N-1)2\pi\sqrt{LC_{\text{filt}}}} \tag{7}$$

where α_{LC} is a user-defined scaling factor. This scaling factor can be used to achieve adequate second-order filter attenuation and to avoid resonance with the LC filter. In this work we chose $f_{\text{sw, min}} = 40$ kHz as an absolute lower limit, which corresponds to a minimum effective switching frequency of $f_{\text{eff}} = 200$ kHz for the 6-level FCML inverter. In this instance, f_{eff} is separated from f_{corner} by $\alpha_{LC} = 3.2$, and the filter attenuates the minimum effective switching frequency $f_{\text{eff}} = 200$ kHz by 19.5 dB.

The full constraints are thus described by

$$f_{\text{sw}}(t) \in [\max(f_{\text{sw, min}}, f_{v_C,\text{ min}}), f_{\text{sw,max}}] \tag{8}$$

and Fig. 4 shows the constrained variable switching frequency $f_{\text{sw}}(t)$ over half of a line cycle for the same FCML converter operating parameters as in Fig. 3. Exaggerated pulse-width modulated gate signals are shown for a small subset of the line cycle, and demonstrate the corresponding reduction in frequency around the low ripple duty cycles. Fig. 5 shows the resulting converter waveforms, and demonstrates that the

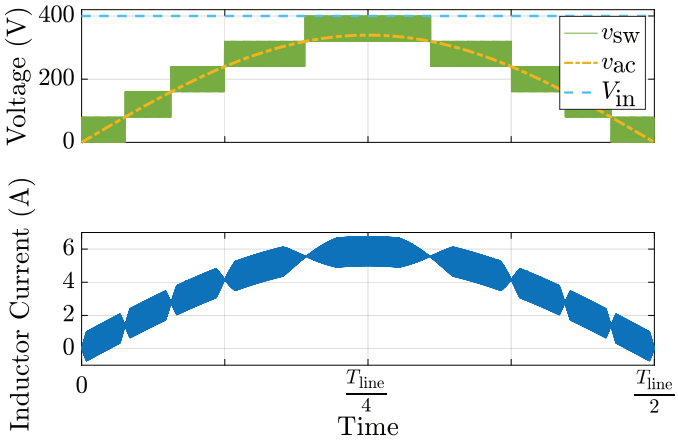


Fig. 5: Calculated waveforms with the proposed technique at $V_{\text{in}} = 400\,\text{V}_{\text{dc}}$, $v_{\text{ac}} = 240\,\text{V}_{\text{rms}}$, $P_{\text{out}} = 1\,\text{kW}$, $C_{\text{fly}} = 3\,\mu\text{F}$, $L = 22\,\mu\text{H}$, $C_{\text{filt}} = 300\,\text{nF}$.

inductor current ripple is largely constant except for small regions near low-ripple duty cycles.

III. Hardware Implementation

The experimental hardware prototype shown in Fig. 6 was validates the variable switching frequency implementation for $V_{\text{in}} = 400\,\text{V}_{\text{dc}}$, and $v_{\text{ac}} = 240\,\text{V}_{\text{rms}}$ up to 1450 W. The inductance of the output inductor $L = 22\,\mu\text{H}$ was chosen to achieve a maximum peak-to-peak current ripple of $\Delta i_{L,\text{pp, max}} = 1.81\,\text{A}$ at a maximum switching frequency of 100 kHz. For the validation, a limit on the flying capacitor voltage ripple is set to $\Delta v_{C\text{fly, max}} = 5.3\,\text{V}$, which corresponds to the peak voltage ripple that would occur at the maximum switching frequency 100 kHz at a full load of 1.45 kW. A less stringent limit would result in even lower switching frequencies across the load range. Converter parameters are described in Table I.

The proposed variable switching frequency technique can be implemented with a digital signal processor (DSP). A TI C2000 MCU is used to generate the PS-PWM signal to control the FCML converter using the rated power level and modulation index D in open-loop. As shown in Fig. 7, the switching frequency is updated at the end of every switching cycle, and the phase shifts and duty cycle comparisons are synchronized concurrently. Careful attention must be paid to ensure proper phase shifting between each switch pair.

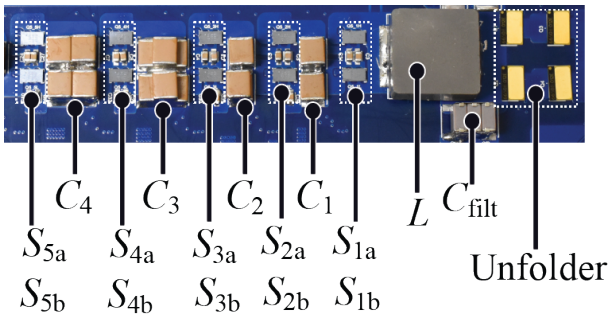


Fig. 6: Annotated photograph of hardware prototype with $C_{1-4} = 3\,\mu\text{F}$, $L = 22\,\mu\text{H}$, $C_{\text{filt}} = 300\,\text{nF}$.

Table I: Converter Operating Conditions

Parameter	Value
V_{in}	400 V
v_{ac}	240 V_{rms}
Measured P_{out}	55 - 1450 W
f_{sw}	40 kHz - 100 kHz
Grid frequency f_{line}	60 Hz
Derated C_{fly}	$\approx 3\,\mu\text{F}$, paralleled $2.2\,\mu\text{F}$ 450 V, C5750X6S225K250KA
FCML GaNFETs	150 V, 7 mΩ, EPC 2033
Unfolder GaNFETs	650 V, 50 mΩ, GS66508T-MR
L	22 µH, IHLP8787MZER220M51
C_{filt}	300 nF, 630 V, CAA573C0G2J304J
Microcontroller	TI C2000 TMDSCNCD28379D

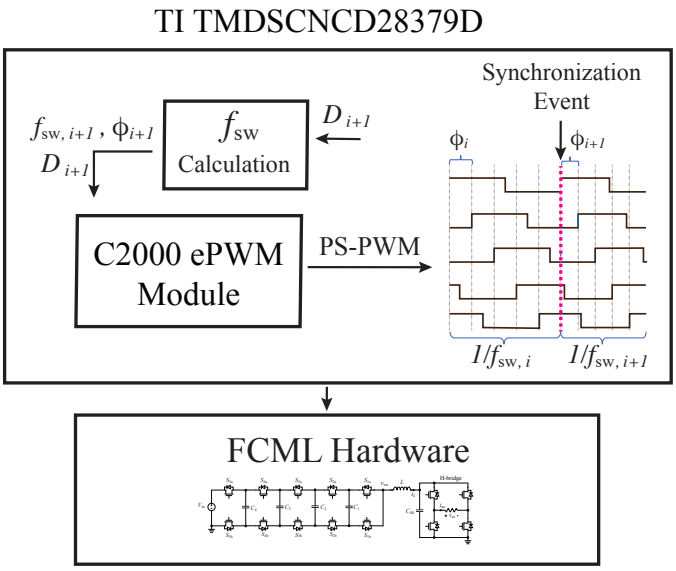


Fig. 7: High-level diagram of micro-controller implementation.

In this work, the switching frequency was updated each switching period. This made it possible to examine the full improvements that may be made by switching at the lowest possible switching frequency while maintaining operational ripple constraints. It may be desirable for other applications to update the switching frequency intermittently along the line cycle to minimize sampling and control complexity.

Fig. 8 shows measured conventional fixed switching frequency operational waveforms at an output power $P_{\text{out}} = 1\,\text{kW}$. The measured inductor current is consistent with the theoretical waveform in Fig. 3. Fig. 9 shows excellent matching between theory and the experimental waveforms for variable switching frequency operation at $V_{\text{in}} = 400\,\text{V}_{\text{dc}}$, $v_{\text{ac}} = 240\,\text{V}_{\text{rms}}$, and an output power of $P_{\text{out}} = 1\,\text{kW}$. Flying capacitor voltage balancing according to the nominal

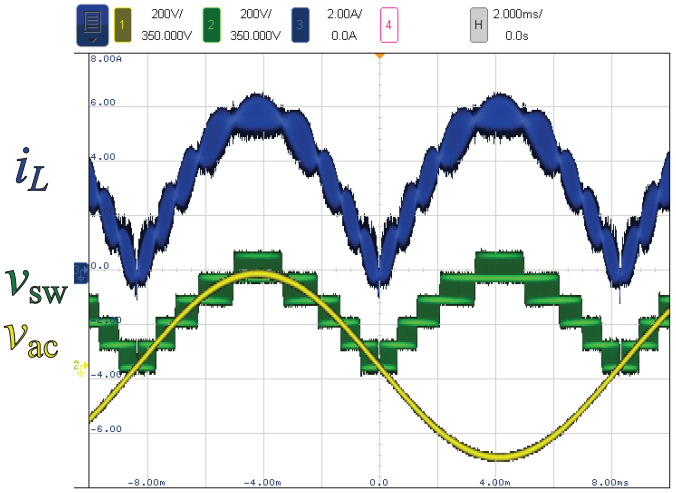


Fig. 8: Oscilloscope plot of i_L, v_{sw}, and v_{ac} for fixed $f_{\text{sw}} = 100\,\text{kHz}$.

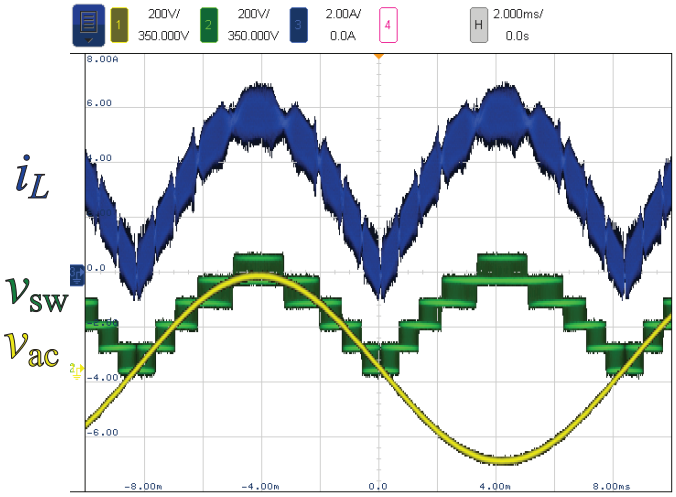


Fig. 9: Oscilloscope plot of i_L, v_{sw}, and v_{ac} for variable $f_{sw} \in [40, 100\,\text{kHz}]$.

distribution is required for converter operation. When the flying capacitor voltages are balanced, the switch node voltage v_{sw} has constant amplitude. For a dc-ac FCML converter, the switch node voltage has distinct voltage steps as the duty cycle varies, but within each step this constant amplitude is maintained. Excellent balancing is observed in the clearly delineated voltage steps in Fig 9. No imbalance in the switch-node voltage waveform was observed at any point across the measured load range.

Most notably, as f_{sw} is reduced during much of the line-cycle, the overall efficiency of the converter improves. Efficiency measurements excluding gate drive losses were taken with a Keysight IntegraVision PA2201A power analyzer. As shown in Fig. 10, the peak efficiency under fixed $f_{sw} = 100\,\text{kHz}$ operation was $98.8\,\%$ at $1.009\,\text{kW}$. Under variable switching frequency operation, the peak efficiency reached $99.03\,\%$ at $562\,\text{W}$. At peak load, 1.45 kW, the efficiency was

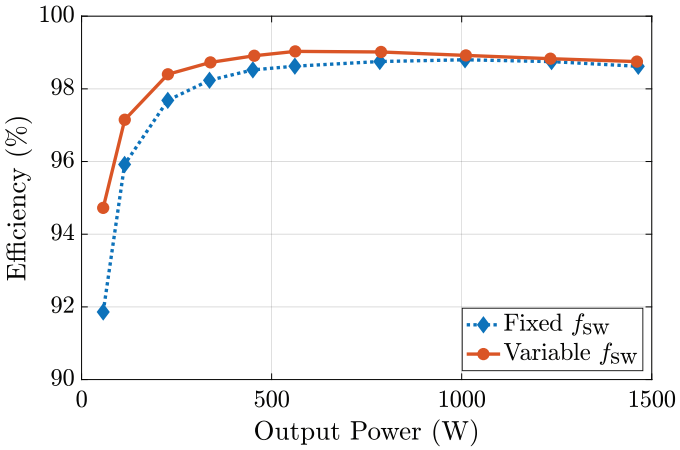


Fig. 10: Plot of efficiencies for both fixed frequency and variable frequency operation.

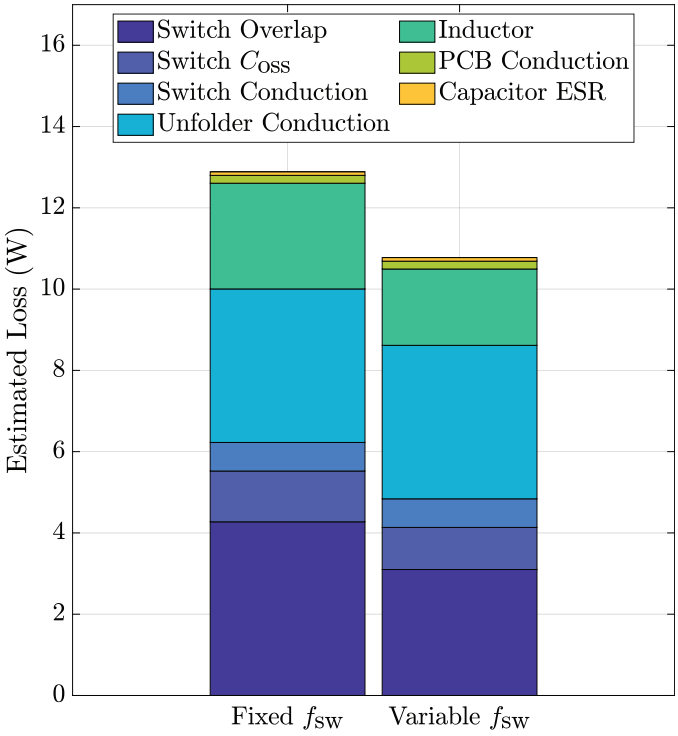


Fig. 11: Estimated loss breakdown for operation at $1\,\text{kW}$.

increased from $98.62\,\%$ to $98.75\,\%$ with the variable switching frequency scheme. This corresponds to a $9.5\,\%$ reduction in losses. At light load the improvements are even more dramatic. At an output power of $60\,\text{W}$, the efficiency increased from $91.86\,\%$ to $94.73\,\%$ which translates to a $35\,\%$ reduction in losses.

An estimated loss breakdown for both operational modes at $1\,\text{kW}$ is provided in Fig. 11, and shows that GaNFET switching overlap losses contribute most drastically to the improvement in efficiency afforded by the proposed variable switching frequency operation. Switching overlap losses were approximated using the sum of the calculated turn-on and turn-off energies at the variable switching frequency across the line cycle. This approximation is simplified as $P_{\text{overlap}} = \frac{2}{T}\int_0^{T/2} v_{ds} \cdot i_{ac} \cdot f_{sw}(t)dt \cdot \frac{t_{\text{turn-on}} + t_{\text{turn-off}}}{2}$ to capture the line-cycle variation in current. Conduction losses throughout the circuit are more simply captured by $P_{\text{cond}} = i^2_{\text{ac, rms}} \cdot R$. Benefits of using a reduced average switching frequency may be less noticeable in high current applications where conduction losses might dominate, and would not be substantially mitigated by an improvement in switching losses.

IV. Conclusion

This work outlines the theory and constraints for a variable frequency switching scheme for FCML inverters to enforce a constant inductor current ripple, along with a description how to implement the theory in practice. The technique can be applied to any FCML dc-ac converter using common control inputs, and it can ultimately be used to drastically

increase converter efficiency by lowering the average switching frequency input to each switch without violating passive component ripple ratings. A 6-level FCML inverter hardware prototype validates the approach and demonstrates significant efficiency improvements over conventional fixed switching frequency techniques. A peak efficiency of $99.03\,\%$ was achieved using the variable switching frequency technique. Efficiency improvements were observed across the measured load range of $60\,\mathrm{W}$ - $1.45\,\mathrm{kW}$, where loss reductions on the order of $10\,\%$ - $35\,\%$ were measured.

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